1.1 Alpha and Beta - Outperfoming the Benchmarks

Defining Alpha and Beta

To show someone just how valuable your insight is when managing a portfolio, you need to understand how alpha and beta work. First, alpha uses a generally accepted definition like this:

> Alpha refers to the excess returns earned by relative-return investors, either above or below the market index to which their performance is benchmarked. When a portfolio outperforms its benchmark index, the percentage return by which the portfolio exceeds the index return will be termed positive alpha. If a portfolio underperforms its benchmark index, we'll say it earned a negative alpha.[1]

Some definitions also spell out the notion of displaying excess returns when adjusting for risk, while others leave that concept embedded within the selection of a benchmark. No matter how you define alpha, it is tightly connected to a related concept: beta.

"Beta is a measure of risk that can apply to an individual stock or to a portfolio of stocks. The average market beta = 1.0. In the case of an individual stock, beta measures how much risk, or volatility, that stock is expected to contribute to the overall volatility of a diversified portfolio."[2]

Alpha measures the percentage by which an investment outperforms its benchmark. An alpha score of 5% means that the investment outperformed the market average by 5% over the period being measured. Alpha takes the beta of the investment into account as well.

We can see this, for example, if we construct two portfolios of five stocks each, selected from the S&P 500®, and using the S&P 500 as a selected benchmark for the portfolios. (Assume the dollar value of the portfolio is divided equally into all five stocks.) It would be a simple thing to compare the alpha and beta measures for both portfolios by using comparison charts. The period under consideration will be the start of May 2024 to the end of October 2024 (see Figure 1.1.1).

FIGURE 1.1.1 Five growth-stock portfolio with SPY as benchmark

Source: tradingview.com

The example in Figure 1.1.1 uses candles to compare the equal weighted portfolio of five stocks: Apple (AAPL), Microsoft (MSFT), Meta (META), Nvidia (NVDA), and Broadcom (AVGO) with State Street's S&P 500 Index ETF (SPY), shown as a line chart.

If we know the beta scores of each stock, we can also calculate the alpha score for the performance of this portfolio. We use the standard formula for alpha:

$\alpha = R_p - (R_f + \beta \times (R_m - R_f))$

where :

α = alpha

β = beta

R_p = return of the portfolio

R_m = return of the benchmark

R_f = risk-free rate (10-year note yield)

The values recorded at the time this chart was captured are as follows:

R_p = return of the portfolio = 34%

R_m = return of the benchmark (SPY) = 17%

R_f = risk-free rate (10-year note yield, TNX) = 4.6% as of the beginning of the chart

β = portfolio beta (average beta score for the five stocks) = 1.24

This allows us to calculate alpha as $\alpha = (.34 - (.046 + 1.24 \times (.17 - .046))) \times 100$, which resolves to an alpha score of 14.02%. That becomes a useful score for comparison purposes, especially if we can use the same benchmark and the same risk-free rate.

For the second portfolio to compare, consider a portfolio of five utility stocks: Duke Power (DUK), Dominion Energy (D), Southern Company (SO), American Electric Power (AEP), and NextEra Energy (NEE).

Comparing the returns of an equal-weighted portfolio of these five utility stocks (dividends included) allows us to see how alpha can be generated even with stocks that have a notably

FIGURE 1.1.2 Five utility-stock portfolio with SPY as benchmark

Source: tradingview.com

One might naturally expect the alpha scores to differ in a similar way. The higher return for the growth stocks over the utility stocks (34% vs. 24.5%) might suggest that the growth stocks generated more alpha than the utility stocks. It is true they did, but only slightly more. The values recorded at the time this chart was captured are as follows:

R_p = return of the portfolio = 24.5%

R_m = return of the benchmark (SPY) = 17%

R_f = risk-free rate (10-year note yield, TNX) = 4.6% as of the beginning of the chart

β = portfolio beta (average beta score for the five stocks) = .53

The utility stock portfolio features stocks with significantly lower beta scores, so the comparison for alpha has a lower starting point. The average score of .53 is well below the beta scores of the growth stocks. That means the returns attributable to market influence (the beta) are much less, allowing the performance to account for more alpha.

Based on these values for the utility stock portfolio, the alpha score calculated as $\alpha = (.245 - (.046 + .53 \times (.17 - .046))) \times 100$, which resolves to an alpha score of 13.32%. In this particular period, even though the utility stocks showed significantly less total gain, they showed almost as much alpha as the growth stocks. It makes for a good demonstration of how calculating alpha can help an investor better gauge a portfolio manager's performance independent of market influence.

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[1] Weigand, Robert A. *Applied Equity Analysis and Portfolio Management: Tools to Analyze and Manage Your Stock Portfolio*. Hoboken, NJ: John Wiley & Sons, 2014.

[2] Ibid.

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Where to Find Alpha

Since alpha is so important, it comes as no surprise that financial professionals spend significant effort seeking where to find sources of alpha. From the perspective of a technical analyst, one would expect to find alpha wherever asset prices are likely to outperform the market averages for significant periods. The quest for alpha becomes an exercise in identifying where trends can be predictably found.

For the technical analyst, that means searching among indicators known for identifying trend breakout and continuation signals. Price tools such as moving averages, oscillators, or price envelope tools like Bollinger Bands, Donchian channels, and Keltner channels are good bets. However, such tools rarely provide persuasive evidence to the more fundamentals-driven investors and professionals in the financial world.

According to proponents of the Efficient Markets Hypothesis (EMH), studying price action is not likely to help someone find predictable sources of new trends that are worth the time and effort to exploit.[3] However, even the academic research community acknowledges it is not impossible.[4] What kind of study should an analyst use to identify trends in a way that could prove persuasive to a client, co-worker, or potential customer?

Pairing reliable technical studies with qualified anomalies, ones known to provide worthwhile challenges to the EMH, is more likely to generate persuasive evidence for any financial or academic professional willing to take their craft seriously. Any worthwhile research effort should take pains to reconcile its findings with one or more of the leading hypothetical market models. Any professional steeped in the guidelines of modern portfolio theory should likewise know to look outside the lines when finding ways to expect the unexpected.

After all, while the EMH is surely the most well known and accepted, it is not the only one, nor is it to be considered an inerrant canon, as Fama himself observed. “It's a model, so it's not completely true. No models are completely true. They are approximations to the world.”[5]

There are at least two other documented market-modeling approaches worth considering for any who would seek a reliable source of alpha, including the Adaptive Market Hypothesis (AMH) and the Fractal Market Hypothesis (FMH).

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[3] Fama, Eugene F., and Kenneth R. French. "Luck versus Skill in the Cross-Section of Mutual Fund Returns." *Journal of Finance* 65, no. 5 (2010): 1915-1947.

"Fact, Fiction, and Momentum Investing." *Journal of Portfolio Management* 40, no. 5 (2014): 75-92. https://dx.doi.org/10.2139/ssrn.2435323.

[5] Fama, Eugene. "Are Markets Efficient?" Moderated by Hal Weitzman. Chicago Booth Review, June 30, 2016. Video, 42:26. https://www.chicagobooth.edu/review/are-markets-efficient.

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The Adaptive Market Hypothesis (AMH)

Author Andrew Lo articulated the AMH, suggesting that financial markets do not efficiently adjust to information because they are made up of investors whose attitudes and behaviors evolve over time. They adapt based on changing expectations and cognitive biases about the markets, and those who adapt best are most likely to survive.[6]

With this framework, Lo suggests that markets may go through periods of informational efficiency and inefficiency. During periods where markets show informational inefficiency, or price behavior better explained by cognitive biases than by new information, adaptations might become necessary for many investors. Lo points out the way for alpha hunters to track and target excess returns in times of known extreme conditions or informational inefficiency.

Volume Patterns at 52-Week Highs and Lows

Although Steven Huddart, Mark H. Lang, and Michelle Yetman did not set out to discover adaptive behavior among investors, their paper “Volume and Price Patterns Around a Stock’s 52-Week Highs and Lows: Theory and Evidence” seems to draw conclusions consistent with the principles Lo articulates in the AMH. They conclude:

> Our results suggest that extreme prices in a stock’s past price path affect investors’ trading decisions in equity markets. Across a broad sample of stocks representative of U.S. equities, volume fluctuates depending on the location of the current price in the distribution of prices over the prior year: volume is higher when the stock price is above the 52-week high or below the 52-week low, suggesting that the prior extremes are salient in decision-making.
>
> While our results do not imply that all investors use these cues, the 52-week highs and lows appear to be salient enough to some investors that this phenomenon can be observed in aggregate volume data. Further, the magnitudes of the effects are statistically significant and are economically as large as or larger than the effects of prominent information events, including earnings announcements and tax-based trading strategies, such as dividend capture. These results are striking because they are associated with an event that does not convey information about firm fundamentals.[7]

The idea that something exploitable occurs as an event solely identified by a unique price for a given equity (a new 52-week markets appear to show informational inefficiency.

When a stock price trades at its highest price, then retraces to lower prices over the course of a year or more, only to trade at a price that exceeds the former high, there should be nothing about this new pinnacle that was not already priced in from former fundamental information. Yet when these researchers identified an increase of trading volume at such a price, having disqualified all other explanations, they surmised that investors are taking action as a result of the new high and no other information.

This kind of investor behavior (or adaptation) is a classic event for which behavioral finance proponents can identify at least one cognitive bias at work. Because investors modify their behaviors in such circumstances, it might become necessary for professional investors to take note of such events and adapt as a way of optimizing their chances for investment survival.

Trading in Pakistan

It is a good thing Fama did not insist that the EMH was nothing but veritas, otherwise Kelly Corbiere's 2022 paper would create problems. In the Journal of Technical Analysis, Dr. Corbiere published an excellent example of targeted research to present a challenge to the EMH that also happened to support the AMH.

The paper studies Pakistan's KSE-100 Index to see if it is weak form efficient, meaning that some degree of predictability can be established from historical price data. Corbiere reports: "The results of statistical tests evaluated in this article suggest that returns in Pakistan's main stock market are not random and not weak form efficient. This implies that investors can generate above-market returns using technical analysis."[8]

It would be hard to make a more dramatic statement of finding when challenging the EMH, but the paper goes on to conclude:

> While the findings are inconsistent with the Efficient Market Hypothesis, they conform to the predictions of the Adaptive Market Hypothesis. Numerous factors may impede the market efficiency, but the EMH requirement for investors to act rationally represents a material hurdle that may not hold for myriad reasons. As this study demonstrated, investors may be slow to incorporate information in a market that is moving strongly in one direction or the other.[9]

This conclusion not only hits the target of the study but offers a useful principle for identifying trends as a bonus. The principle may not offend the EMH model, but it is certainly consistent with the AMH framework. It can be summarized this way: Strong trends in markets that more slowly incorporate new information are more likely to continue rather than reverse. This simple but powerful principle forms a foundation for a technical analyst to build new research on and find strategies for harvesting alpha in untapped sources.

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[6] Lo, Andrew W. *Adaptive Markets: Financial Evolution at the Speed of Thought*. Princeton, NJ: Princeton University Press, 2017.

[7] Huddart, Steven J., Mark H. Lang, and Michelle Yetman. "Volume and Price Patterns Around a Stock's 52-Week Highs and Lows: Theory and Evidence."

Market Efficiency in Pakistan." *Journal of Technical Analysis* 72 (2022): 45-68. https://cmtassociation.org/journal_ta/issue-72/

[9] Ibid.

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The Fractal Market Hypothesis (FMH)

Author Edgar Peters is a 40-year practitioner in asset management who articulated the FMH in the 1994 publication of his book *Chaos and Order in the Capital Markets*.[10] He wrote the book in part with the ambition of explaining investor behavior in periods of crisis. Four notable bear markets have taken place in the three decades since, and Peters believes the FMH explains investor behavior in each of them.

Peters argues that differing decision horizons and risk tolerance among investors can lead to a wide variety of reactions. These include reactions to market events both fundamental and technical in nature. He further argues that it is not enough for EMH proponents to lump together all historical price data into a single category. Investors do not make decisions based on an aggregate of all historical prices, but rather on differing information sets.

Peters maintains that it is the combination of diverse investor behaviors and variable information sets that leads to periods of market uncertainty. Naturally, when enough investors perceive the market in dissimilar ways and, further, they disagree on what to do about it, price instability and volatility can increase.[11] Without necessarily intending to do so, Peters's proposition of the FMH establishes the possibility of a much more aggressive rejection of the EMH. Most of its conclusions lead to the possibility of predicting future price action in some degree.

Peters's claims about the FMH have been taken up by some researchers that do offer corroborating evidence for the hypothesis. Among the more interesting offerings, one looks at wavelet power spectra as a way of analyzing stock markets.[12] Another uses a Markov regime switching model (a useful statistical tool for classifying binary state changes) to identify high and low risk periods in stock prices.[13]

These studies suggest that the FMH has merit to be considered by offering corroborating evidence in applications of limited scope. More research on the FMH is warranted, building on the foundation laid by others[14] — in particular, verifying or rejecting the FMH notion that prices can be memory associative. This notion implies that not only are future prices data dependent on recent information but also on the cumulative data built into all historical price data for an asset.

Additional research for this market model would simultaneously support the possibility for some price data to be predictive as

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[10] Peters, Edgar E. *Simple and Complex Market Inefficiencies: Integrating Efficient Markets, Behavioral Finance, and Complexity."* Journal of Behavioral Finance 4, no. 1 (2003): 22-35. https://www.researchgate.net/publication/247504663_Simple_and_Complex_Market_Inefficiencies_

[11] Peters, Edgar E. *Fractal Market Cycles and Regimes*. New York: Wiley, 2022.

[12] Kristoufek, Ladislav. "Fractal Markets Hypothesis and the Global Financial Crisis: Wavelet Power Evidence." *Scientific Reports* 3 (2013): 1848-1861. https://doi.org/10.48550/arXiv.1310.1446.

[13] Doorasamy, Mishelle, and Prince Kwasi Sarpong. "Fractal Market Hypothesis and Markov Regime Switching Model: A Possible Synthesis and Integration." *International Journal of Economics and Finance* 8, no. 1 (2018): 93-100. https://ssrn.com/abstract=3111614.

[14] Blackledge, Jonathan, and Marc Lamphiere. "A Review of the Fractal Market Hypothesis for Trading and Market Price Prediction." *Mathematics* 10, no. 1 (2022): 117. https://doi.org/10.3390/math10010117.

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Supply and Demand

Technical analysts find discussing supply and demand to be a simple thing. For example, when Edson Gould opined in his newsletter circa 1977 that "the utilities [sector] reflects to a greater extent than the Industrials the investment demand for stock,"[15] it was a simple concept to discuss, but much harder to precisely measure.

The phrase "investor demand" implies an overall willingness to buy securities, so much so that the resulting orders are likely to increase the price of the securities in question. This notion is general to any kind of securities, including stocks, bonds, commodities, currencies, cryptocurrency, and so forth. But exactly how much demand is enough to move the price higher? Exactly what trading volume of initiated buy orders signifies not only demand, but the persistence of demand in the near future?

By contrast, the concept of supply implies that investors have a quantity of securities they prefer to sell, so much so that they are likely to drive the price lower in the process. However, as with demand, the finer details of cause, effect, and necessary quantities have eluded definition.

Despite this significant gap in between concept and causation, the general notion of supply and demand for securities seems to persist. Perhaps because it makes a readily available mental construct that investors and traders can use to interpret what they are seeing in the markets — venturing into speculation about what might come next. Those who do would be wise to consider coupling observations of supply and demand with known anomalies the EMH cannot explain.

Consider the following list of some of the more prominently known anomalies and their suggested relationship to supply and demand. Perhaps the combinations of these ideas will generate productive research for capturing alpha.

Known market anomalies not explained by the EMH	Results that go against EMH predictions	Explanations and opportunities
Correlations of returns in groups of related securities	Prices are not adjusting efficiently, and investors may be able to predict a trend could continue.	Investors show supply or demand for similar securities.

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explained by the EMH	predictions	opportunities
Mean reversion	Investors may be able to predict that portfolios with the highest returns underperform, while those with worst returns outperform.	The desire to capture returns creates supply, the desire to redeploy the cash creates demand for undervalued shares.
Underreaction to news	Prices may predictably continue in trends following both good and bad news.	Supply and demand have more influence at times, and investors may consider the news unimportant.
The size effect	Investors may correctly predict that small stocks outperform large stocks, even accounting for higher volatility.	Relative volume increases may trigger a recognition of improved volume.
The P/E ratio (value investing)	Lower P/E scores tend to predict greater outperformance compared to other stocks.	Low demand (or higher supply) for stocks over longer periods creates underpriced stocks.
The January effect	Small cap stocks outperform early in the year.	High demand for these stocks early in the year creates opportunities to watch for.
Patterns in volatility	Bear markets are more volatile than bull markets; the opening and closing minutes have more volatility than the rest of the day.	Excess supply creates volatility, and volatility from excess demand is more likely to dissipate quickly.

[15] Gould, Edson. "The Utility Barometer: An Early Stock Market Indicator." *Edson Gould's "Findings & Forecasts" Newsletter*, 1974.

[16] Weigand, Robert A. *Applied Equity Analysis and Portfolio Management: Tools to Analyze and Manage Your Stock Portfolio*. Hoboken, NJ: Wiley, 2014.

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Conclusion

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Conclusion

Technical analysts and researchers would do well to thoroughly understand the EMH and its challenges, specifically the AMH and FMH, among others. The anomalies listed previously are among those used to challenge the EMH and are an excellent starting point for researchers. Using indicators such as moving averages, oscillators, and price envelope tools during times when these anomalies present themselves are more likely to generate reliable sources of alpha.

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